

Supplementary Information for: Examining the Relational Content and Personal Determinants of Social Signatures Across Multiple Time Scales

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Supplementary Table S1: Per-Channel Robustness Comparison

The main text pools all outgoing communication events (voice calls, SMS, MMS, and WhatsApp messages) into a single, equal-weighted measure of communication effort, a simplification discussed in the Limitations section. Table S1 reports the headline results separately for the voice-call channel alone, the combined text-based channels alone (SMS, MMS, and WhatsApp), and the pooled measure used in the main text, using identical semester-window definitions, eligibility criteria (≥ 2 alters and > 10 channel-specific events per window), and data-quality exclusions throughout.

Table S1: Per-Channel Robustness Comparison: Calls-Only vs. Texts-Only vs. Pooled (Semester Windows)

Metric	Calls-Only	Texts-Only (SMS+MMS+WhatsApp)	Pooled (Main Text)
Eligible Egos (N)	285	283	307
Mean Self-Divergence (d_{self})	0.062	0.066	0.066
Mean Reference-Divergence (d_{ref})	0.120	0.139	0.132
Wilcoxon Signed-Rank Test (p , $d_{\text{self}} < d_{\text{ref}}$)	< 0.0001	< 0.0001	< 0.0001
Mean Power-Law Exponent (α)	1.18	1.94	1.94
Power-Law Preferred over Exponential by AIC (%)	96.6%	38.2%	60.8%
Rank 1 Alter is Family (%)	72.1%	19.5%	23.3%
Rank 1 Alter is Friend (%)	16.0%	56.1%	53.1%
Rank 1 Alter Retained Across Semesters (%)	59.9%	41.0%	40.5%
Rank 1 Dyads Matched to Survey (N)	933	912	1,003

Interpretation. Two conclusions follow from this comparison. First, the core persistence finding—that intraindividual self-divergence is reliably smaller than interindividual reference-divergence—is channel-invariant: it holds with $p < 0.0001$ whether the signature is built from calls alone, texts alone, or the pooled measure, confirming that signature stability is not an artifact of the equal-weighting assumption or of any single channel’s measurement properties. Second, the channel-specific compositional and curvature findings reported in the main text are not channel-invariant. The voice-call channel alone reproduces a kinship-dominated, power-law-preferred, more stable primary slot (72.1% family at Rank 1, 96.6% power-law preference, 59.9% top-alter retention), closely matching the calls-only results reported in earlier work using this cohort. The text-based channels

alone instead produce a friend-dominated, more volatile primary slot with a steeper decay exponent and weaker power-law preference (19.5% family, 56.1% friend, 38.2% power-law preference, 41.0% retention). Because text messages constitute 97.7% of all logged NetHealth communication events, the pooled measure used in the main text closely tracks the text-only channel on every metric (e.g., $\alpha = 1.94$ for both texts-only and pooled), while the comparatively rare but higher-investment voice-call channel is numerically diluted. This confirms the main text’s interpretation that the family-to-friend compositional reversal documented for the pooled Rank 1 slot is substantially attributable to the equal per-event weighting of a high-volume, low-marginal-cost channel (texting) alongside a low-volume, high-investment channel (calling), rather than reflecting a change in the underlying persistence or slot-filling mechanisms, both of which replicate across all three channel definitions.

Supplementary Table S2: Statistical Tests of Signature Persistence

Table S2 reports the full statistical comparison between intraindividual self-divergence (d_{self}) and interindividual reference-divergence (d_{ref}) across all eight temporal window schemes evaluated in the main text (Section 4.2).

Table S2: Statistical Tests of Social Signature Persistence Across Window Schemes

Scheme	Egos (N)	Mean d_{self} (SD)	Mean d_{ref} (SD)	Difference	Wilcoxon V	p -value
Academic Years	405	0.0584 (0.0640)	0.1255 (0.0464)	−0.0671	7,151.0	< 0.0001
Semesters	307	0.0658 (0.0630)	0.1324 (0.0437)	−0.0666	3,602.0	< 0.0001
Quarters	259	0.0425 (0.0264)	0.1342 (0.0418)	−0.0918	0.0	< 0.0001
Months	208	0.0414 (0.0189)	0.1369 (0.0359)	−0.0954	0.0	< 0.0001
Common Cohort (Sem)	208	0.0582 (0.0559)	0.1260 (0.0439)	−0.0678	1,419.0	< 0.0001
Common Cohort (Mo)	208	0.0414 (0.0189)	0.1369 (0.0359)	−0.0954	0.0	< 0.0001
3-Week Rolling	187	0.0182 (0.0077)	0.1379 (0.0353)	−0.1197	0.0	< 0.0001
2-Week Discrete	180	0.0450 (0.0169)	0.1388 (0.0321)	−0.0938	0.0	< 0.0001

Supplementary Figure S1: Parameter Burn-In Convergence

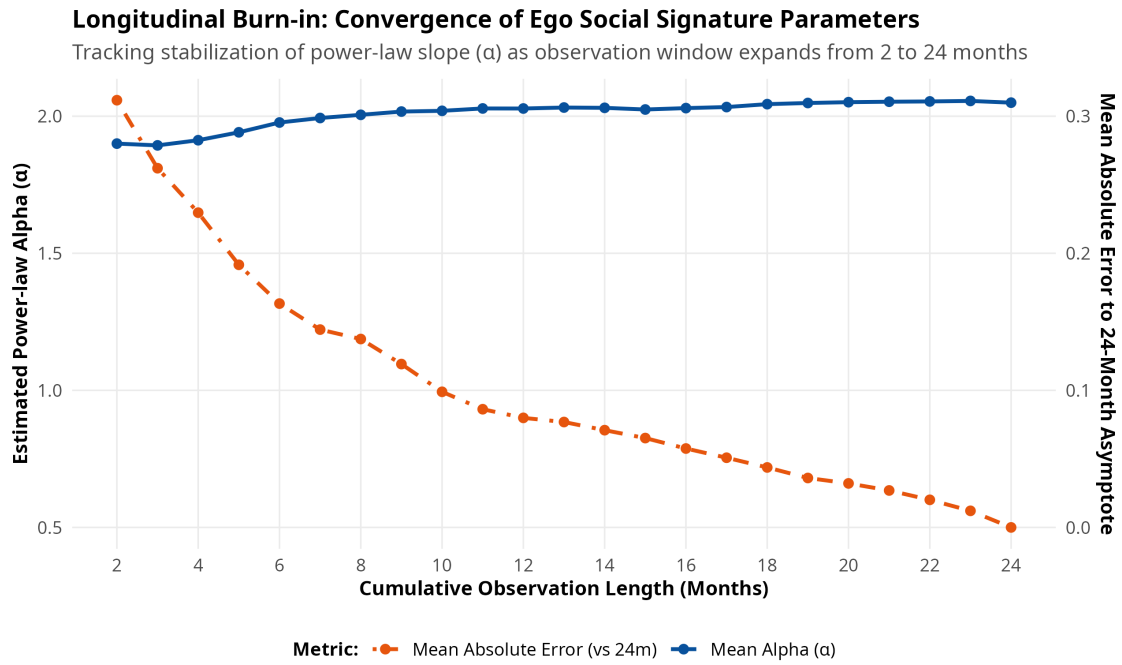


Figure S1: Parameter Burn-In Convergence Over 24 Months. Left axis shows mean estimated power-law α ; right axis shows mean absolute error relative to the 24-month asymptotic parameter value. See main text Section 4.3.

Supplementary Table S3: Parametric Model Evaluation

Table S3 reports the full power-law versus exponential model comparison underlying the summary statistics in main text Section 4.3.

Table S3: Parametric Model Evaluation Across Window Resolutions

Scheme	Models (N)	Power-Law α (SD)	Power-Law R^2	Exp β	Exp R^2	% PL Preferred
Semesters	1,226	1.94 (0.33)	0.933	0.08	0.900	60.8%
Months	4,985	1.77 (0.43)	0.912	0.19	0.911	47.8%

Supplementary Figure S2: Dimensions of Social Support Across Signature Rank Tiers

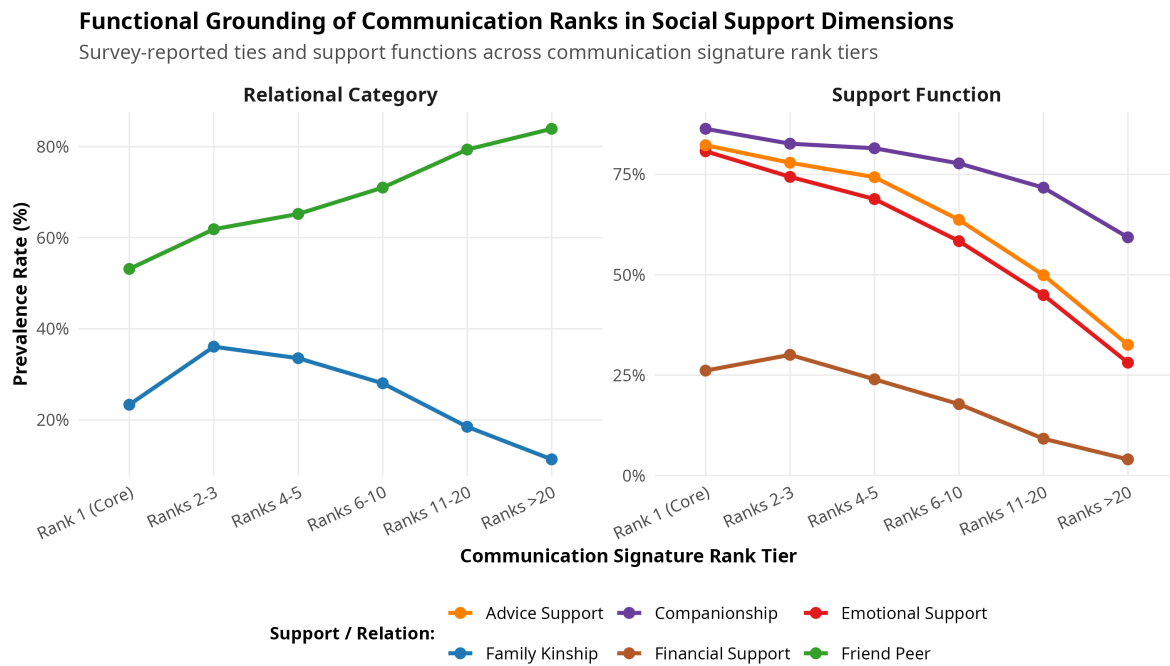


Figure S2: Dimensions of Social Support Across Signature Rank Tiers. Shows prevalence rates of relational categories (left panel) and substantive social support provisions (right panel), visualizing the data reported in main text Table 2.

Supplementary Table S4: Tie-Attribute Regressions

Table S4 reports the full linear mixed-effects models predicting signature allocation and rank from tie attributes, underlying the summary reported in main text Section 5.2.

Table S4: Linear Mixed-Effects Models Predicting Signature Allocation and Rank from Tie Attributes ($N = 16,496$ Dyad-Window Observations across 295 Egos)

Predictor Term	Model 1 (Evaluative)	Model 2 (+ Duration)	Model 3 (+ Salience)	Model 4 (Full Rank)
Intercept	−0.124*** (0.0061)	−0.143*** (0.0064)	−0.160*** (0.0062)	67.5*** (1.36)
Subjective Closeness (1–4)	0.0349*** (0.0020)	0.0389*** (0.0020)	0.0237*** (0.0021)	−6.19*** (0.45)
Interpersonal Trust (1–10)	0.0059*** (0.0007)	0.0076*** (0.0007)	0.0066*** (0.0007)	−0.376* (0.15)
Tie Duration (Years)	–	−0.0013*** (0.0001)	−0.0013*** (0.0001)	−0.222*** (0.03)
Cognitive Salience (26 – Order)	–	–	0.0038*** (0.0002)	−1.09*** (0.04)
Dependent Variable	Comm. Proportion (p)	Comm. Proportion (p)	Comm. Proportion (p)	Signature Rank (r)
Dyad-Window Observations	16,496	16,496	16,496	16,496
Ego Clusters	295	295	295	295
Ego Random Variance (τ_{00})	0.00063	0.00077	0.00047	29.77
Residual Variance (σ^2)	0.00917	0.00908	0.00889	411.69
Intraclass Correlation (ICC)	0.065	0.079	0.051	0.067

Standard errors in parentheses. *** $p < 0.0001$, ** $p < 0.01$, * $p < 0.05$.

Supplementary Table S5: Multilevel Models of Signature Divergence

Table S5 reports the full multilevel linear mixed-effects models predicting signature self-divergence, underlying the summary reported in main text Section 5.3.

Table S5: Multilevel Linear Mixed-Effects Models Predicting Signature Self-Divergence (JSD)

Predictor Term	Model 1 (Turnover)	Model 2 (Topology)	Model 3 (Integrated)
Intercept	−0.207*** (0.0227)	−0.210*** (0.0279)	−0.170*** (0.0430)
Alter Turnover (1−Jaccard)	0.371*** (0.0306)	0.346*** (0.0301)	0.349*** (0.0370)
% Activity Change ($ \Delta\text{Events} /\text{Events}$)	—	0.00027*** (0.00004)	0.00020*** (0.00005)
Ego Network Clustering Coefficient	—	0.023 (0.023)	0.041 [†] (0.023)
Ego Network Density	—	0.002 (0.023)	—
Baseline Extraversion	—	—	−0.0048 (0.0054)
Baseline Negative Emotionality	—	—	−0.0180* (0.0072)
Observations	921	915	642
Ego Groups	307	307	307

Standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, [†] $p < 0.10$.

Supplementary Table S6: Personality Determinants of Signature Shape

Table S6 reports the full OLS regression of ego-mean power-law decay exponents on Big Five personality traits, underlying the summary reported in main text Section 5.4.

Table S6: Personality Determinants of Social Signature Power-Law Alpha (α_i)

Term	Estimate	Std. Error	<i>t</i> value	<i>p</i> -value
(Intercept)	1.973***	0.194	10.17	< 0.0001
Extraversion	−0.026	0.021	−1.22	0.224
Negative Emotionality	0.076**	0.024	3.19	0.0016
Agreeableness	−0.014	0.031	−0.44	0.659
Conscientiousness	−0.003	0.028	−0.09	0.927
Openness	−0.034	0.034	−1.01	0.314
Personal Network Degree	0.0001	0.003	0.04	0.969

$N = 304$ egos, $R^2 = 0.058$, $F(6, 297) = 3.05$, $p = 0.0065$.

Supplementary Figure S3: CART Decision Tree

CART: Predicting Signature Alpha from Big Five Personality

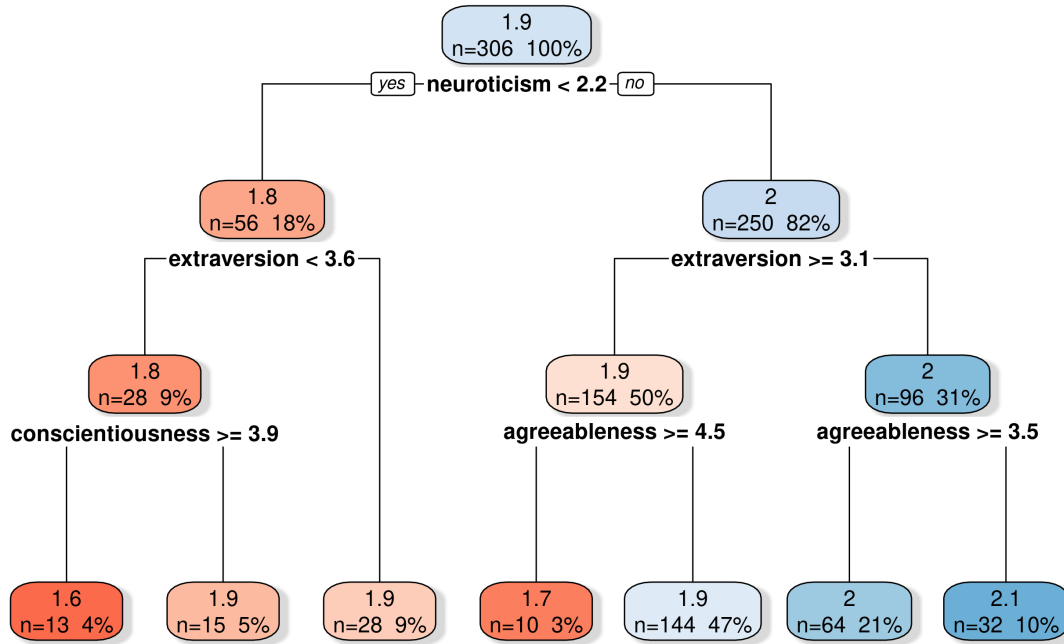


Figure S3: Classification and Regression Tree (CART) Predicting Social Signature Power-Law Alpha (α_i) from Big Five Personality Traits ($N = 306$). Node boxes display predicted mean α_i (top) and sample size (bottom). Cross-validated error exceeds 1.0 at every split, indicating weak out-of-sample predictive value; the tree is reported as an exploratory, descriptive structure only. See main text Section 5.4.